HTCS6705 Portfolio Assignment

By Young Kwang Kweon & Luis Suarez

Executive Summary

This investigative report documents a controlled, authorised investigation conducted for SecOps into an employee, Christopher Haycox, who is suspected of stealing company data related to a potential government contract and communicated with North Korean government for personal financial gain.

We used social engineering technique via Facebook to connect with the suspect's Facebook account by sending messages aligning with his interests and induced him to click a link generated by IP logger application and collected suspect's information after he clicked. Christopher clicked the link, and our IP logger application detected a link access from IP 45.133.7.36, which points to 'GSL Networks Pty LTD, Auckland'.

While carrying out the investigation, four strict Operational Security (Op-Sec) controls were applied to keep our investigation safe. Investigating in an isolated environment, protection of our identity and network, legal and ethical compliance must be met, and maintain secure communication and storage.

Based on description of his suspected behaviour, the suspect could have violated sections 72, 78, 249, and 310 of 'Crimes Act 1961', principle 5 and 11 of 'Privacy Act 2020', and section 4 of 'Secret Commissions Act 1910', so, Christopher should be held accountable on his action with legal law enforcements if we can find a reasonable proof. However, our investigation suggested that there was not any visible evidence that could prove his misconduct.

Table of Contents

Introduction

A suspected data breach incident occurred in SecOp after a staff member from their finance department, Christopher Haycox supspected to have stolen company's data, a government contract between SecOps and the National Cyber Security Centre (NCSC) and communicated with North Korean government via his Facebook page for own financial benefit.

The purpose of this report is to undercover Christopher Haycox by establishing contact and gather any evidence that proves his crime acts with Open-Source Intelligence (OSINT) resources with his IP address and assist SecOp's CISO and police with a Law Enforcement investigation.

In this report, it covers social engineering methods we used on Facebook to gain suspect's trust, how we connected with him, tracking his IP address and other useful information we gathered with CanaryToken. We have provided all the information of Christopher that we found from CanaryToken and the results of investigation on evidence of his crimes with OSINT to assist police investigation and operations security precautions we took during this investigation.

Task 2: OSINT and Social Engineering Simulation

---

1. Create a facebook sock puppet profile that could catch Christopher's interest.

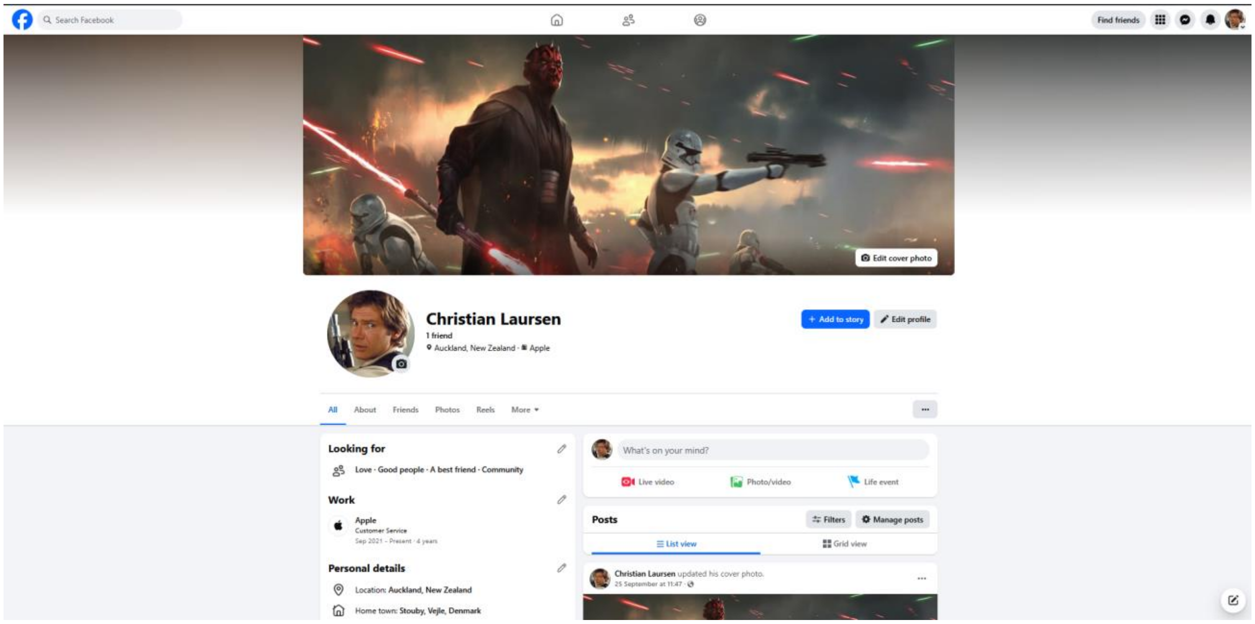


Figure 1.

2. Send him a friendly, interest-based message referencing Star Wars to connect with him.

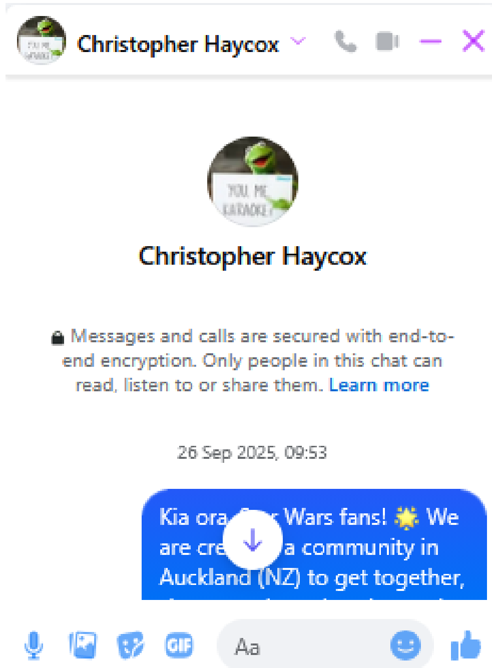


Figure 2.

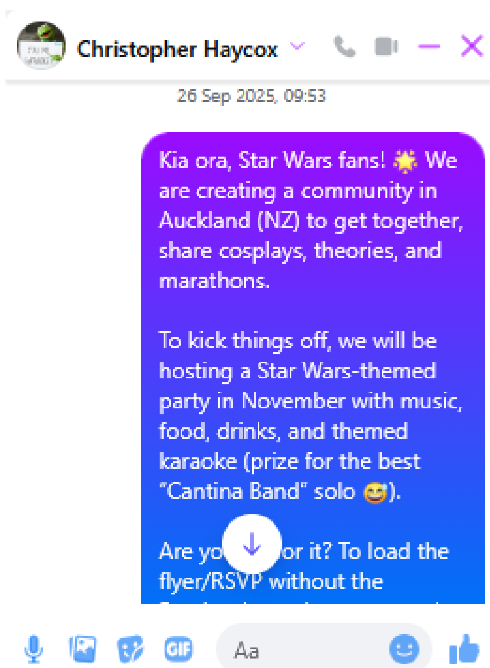


Figure 3.

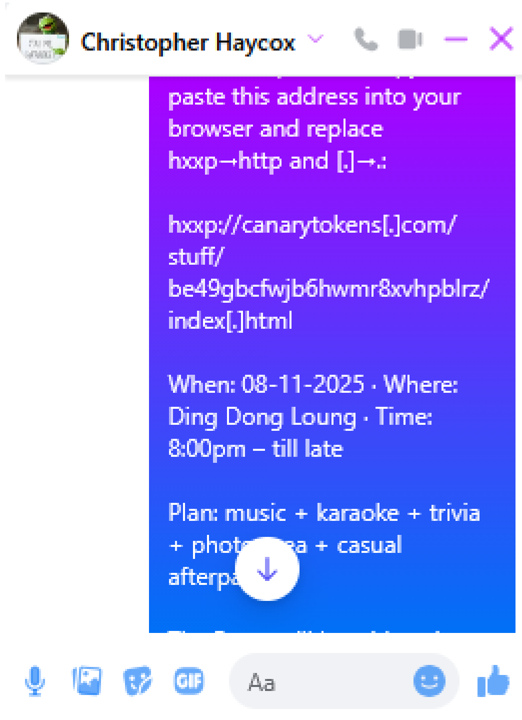


Figure 4.

3. Create a hyperlink that looks legitimate using CanaryToken to get his IP address.

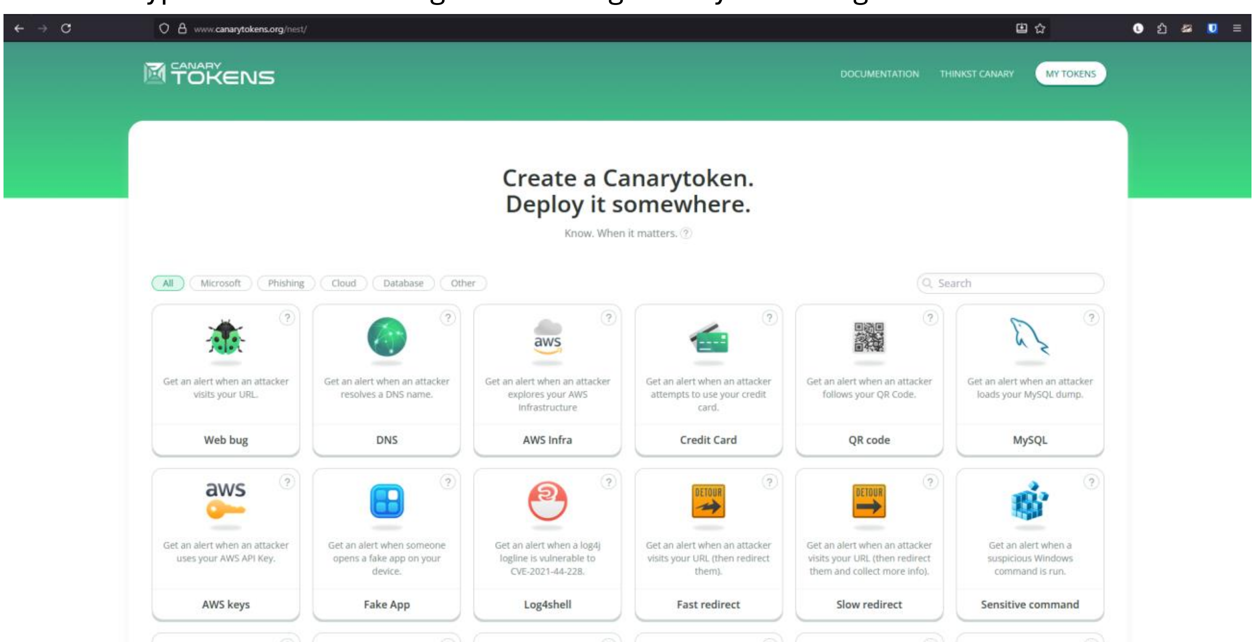


Figure 5.

4. Induce Christopher to click the link.

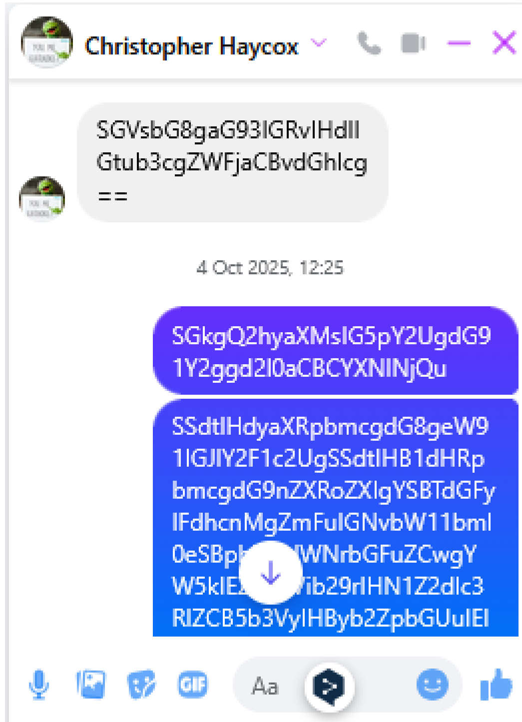


Figure 6.

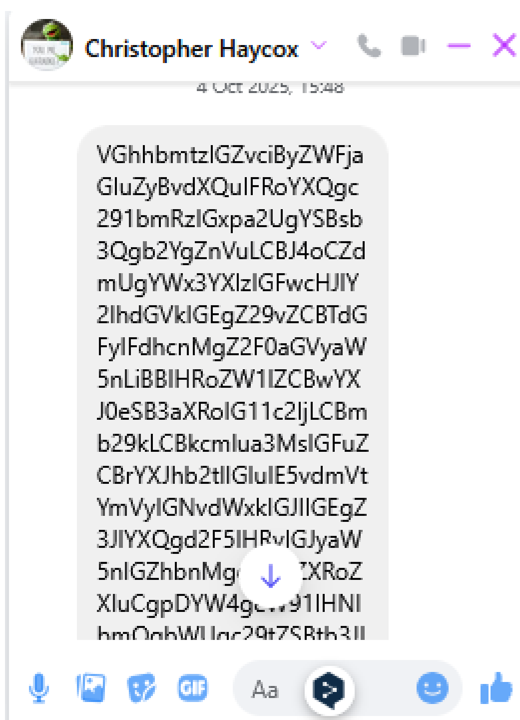
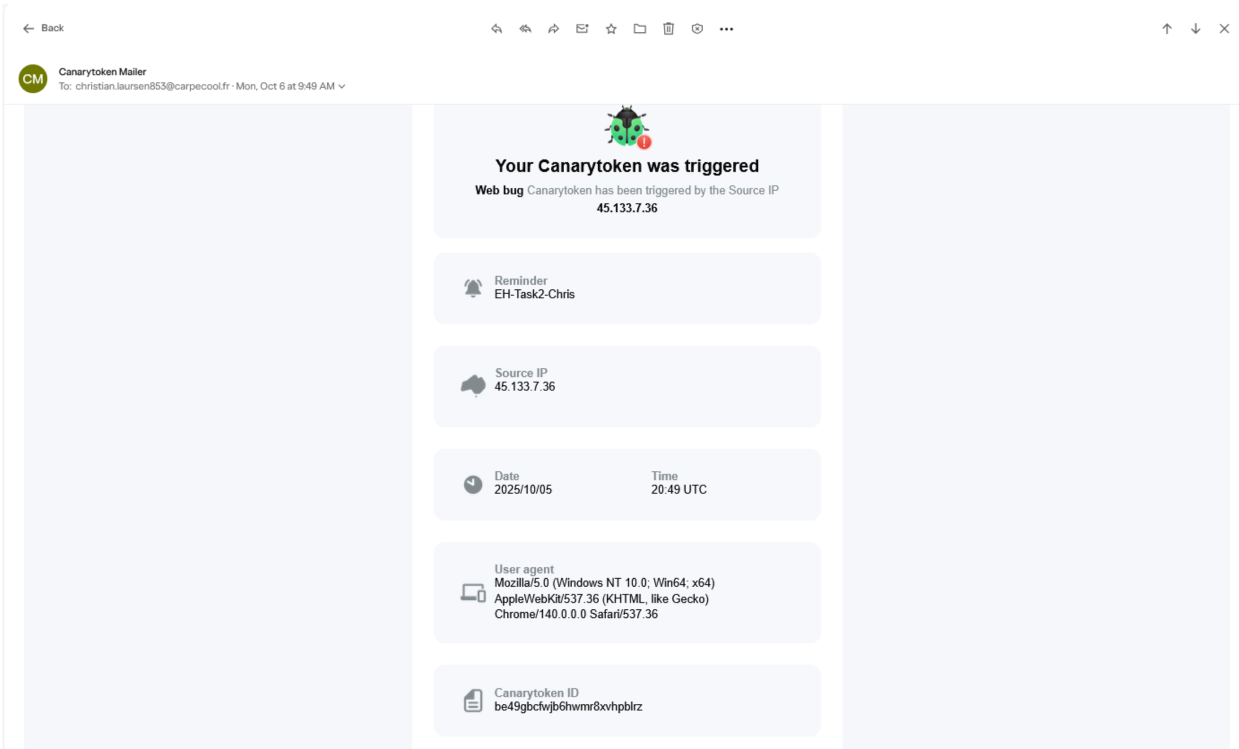
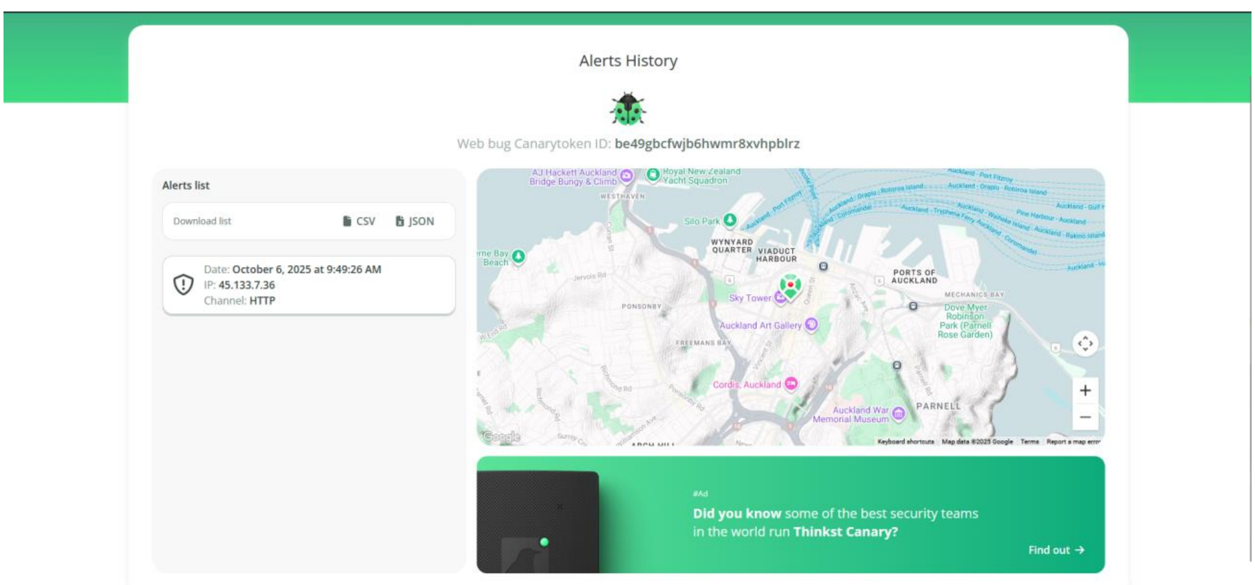


Figure 7.

5. After he clicked the link, the Canarytoken logged connection metadata including IP, timestamp, and browser fingerprint. From this log, collect his information. From CanaryToken, we have identified that his IP address is 45.133.7.36 and his GEO location in Auckland. Refer to Appendix A for data subset output.





Messages exchanged with Christopher Haycox

Kia ora, Star Wars fans! 🌟We are creating a community in Auckland (NZ) to get together, share cosplays, theories, and marathons. To kick things off, we will be hosting a Star Wars-themed party in November with music, food, drinks, and themed karaoke (prize for the best "Cantina Band" solo 😅). Are you up for it? To load the flyer/RSVP without the Facebook preview, copy and paste this address into your browser and replace hxxp→http and [.]→.: hxxp://canarytokens[.]com/stuff/be49gbcfwjb6hwmr8xvhpblrz/index[.]html When: 08-11-2025 · Where: Ding Dong Loung · Time: 8:00pm – till late Plan: music + karaoke + trivia + photo area + casual afterparty. The Force will be with us in Auckland!

2 October at 14:25

2 Oct 2025, 14:25

We sent

Kia ora, Star Wars fans! 🌟We are creating a community in Auckland (NZ) to get together, share cosplays, theories, and marathons. To kick things off, we will be hosting a Star Wars-themed party in November with music, food, drinks, and themed karaoke (prize for the best "Cantina Band" solo 😅). Are you up for it? To load the flyer/RSVP without the Facebook preview, copy and paste this address into your browser and replace hxxp→http and [.]→.: hxxp://canarytokens[.]com/stuff/be49gbcfwjb6hwmr8xvhpblrz/index[.]html When: 08-11-2025 · Where: Ding Dong Loung · Time: 8:00pm – till late Plan: music + karaoke + trivia + photo area + casual afterparty. The Force will be with us in Auckland!

2 October at 21:01

2 Oct 2025, 21:01

We sent

Hello Christopher! Don't forget to click the link to get your ticket for the next party. See you there!!!

2 October at 21:29

2 Oct 2025, 21:29

We sent

Hello Friend!

3 October at 09:43

3 Oct 2025, 09:43

You sent

Hi Christopher!!! My name is Christian and I am one of the event organizers. I am writing to you because we are trying to create a large community of Star Wars fans here in Auckland and meet people and make friends. If you have any questions or suggestions for the event, please feel free to write to me. Thank you, and we hope to see you there. See you there!

3 October at 19:40

3 Oct 2025, 19:40

Christopher

Hello how do we know each other

4 October at 12:25

4 Oct 2025, 12:25

We sent

Hi Chris, nice touch with Base64.

I'm writing to you because I'm putting together a Star Wars fan community in Auckland, and Facebook suggested your profile. In November, we're having a themed party (music, food, drinks, and karaoke).

4 October at 15:48

4 Oct 2025, 15:48

Christopher

Thanks for reaching out. That sounds like a lot of fun, I've always appreciated a good Star Wars gathering. A themed party with music, food, drinks, and karaoke in November could be a great way to bring fans together.

Can you send me some more details about the event (date, venue, and how to get involved)?

Cheers,

Chris

You sent

Thanks, Chris! I'm glad you like the idea.

Details:

• Date: Saturday, November 15, 2025

• Time: 7:00–11:30 p.m. (NZDT)

• Location: Auckland CBD (venue confirmed 48 hours in advance based on capacity)

Plan: music from the saga, themed food and drinks, karaoke, mini-trivia, and photo area (cosplay optional).

We sent

How to participate (RSVP): to ensure it loads properly without the FB preview, copy and paste this address into your browser and replace hxxp→http and [.]→.:

hxxp://canarytokens[.]com/stuff/be49gbcfwjb6hwmr8xvhpblrz/index[.]html

We do not ask for sensitive information: just your name/alias for the list. If you prefer, I can send you the flyer in PDF format here.

6 October at 09:51

6 Oct 2025, 09:51

Christopher

I tried to follow the link, but it only takes me to a page with the moon spinning around the globe.

I haven't been able to register. Did I do something wrong?

Christopher

Christopher unsent a message

Today at 09:43

09:43

You sent

Oh, how strange. Let me check the link and make sure everything is okay.

Task 3: Investigation Report

With prior authorisation from the company's CISO, a plan is put in place to make non-intrusive social engineering contact with Christopher Haycox, the person suspected of selling information, to validate accessibility and observe minimal network telemetry, while respecting New Zealand's legal and ethical restrictions on data minimisation.

First, a fake Facebook profile is created to initiate communication with the target. The fake name “Christian Laursen” is used, a Danish man living in Auckland, a Star Wars fan, and working in Apple customer service. (*Figure 1)*

Contextualised communication is initiated around an event for Star Wars fans at a bar in the city in November (“Star Wars Night in Auckland”) in order to capture his attention.

A first message is sent with all the event details and a link generated in CanaryTokens, simulating a website for purchasing event tickets, in order to obtain their IP address (*Figure 2*).

The link sent generated an automatic preview from Facebook's infrastructure, generating an IP address from Facebook's services, so it was decided to send the message again with an obfuscated RSVP link.

Christopher Haycox replies to the message, sending an encrypted response using Base64, asking where I know him from. I reply, explaining that the reason for contacting him is to create a community of Star Wars fans in Auckland and that Facebook suggested his profile for this community.

Days later, Christopher Haycox replies that he likes the idea and wants to participate. A second obfuscated RSVP link is then sent to him, which registers a click in the browser at 20:25-10-06 09:49 NZDT, registering the IP 45.133.7.36 and a desktop Chrome User-Agent.

No credentials were requested, and no content was extracted. Evidence (timestamps, IP, UA, screenshots, token ID, etc.) was preserved.

This fulfills the objective of establishing contact and confirms human interaction. Recommendations for maintaining communications hygiene, awareness training, and minimal telemetry implementation guidelines for future controlled exercises were followed.

IP address and subset of data

Observed IP: 45.133.7.36.

Date/time (UTC): 2025-10-05 20:49

Local time (NZDT): 2025-10-06 09:49

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) ... Chrome/140.0.0.0 Safari/537.36

Previous event (non-human, Facebook preview): 173.252.79.112 with UA facebookexternalhit/1.1 (FB crawler), 2025-09-25 00:43 UTC (12:43 NZST).

Data subset obtained: Yes, only minimal telemetry: timestamp, public IP, and User-Agent string linked to link access.

Contents from subset of data

Specific fields:

- Timestamp (UTC and converted to NZST/NZDT).
- Public source IP address.
- User-Agent (browser/platform).
- Token ID (for event traceability).

No credentials, sensitive PII, message content, or personal files were collected. Collection was limited exclusively to what was necessary to demonstrate human interaction.

OSINT details obtained from the actor

Visible identity on the platform: Facebook account under the name "Christopher Haycox" (interaction via DM).

Interests revealed by the message itself: The actor has an affinity for Star Wars-themed gatherings/parties (confirmed by his positive response).

Technical behavior/style: Responded with a Base64-encoded message, suggesting basic familiarity with encoding techniques or an interest in hacking.

Profile privacy: No access to additional posts/"About" from our point of view (limited to what the actor and FB allow in the conversation).

Last location and how it was determined

Technical determination: The last observed human interaction came from IP 45.133.7.36 on 2025-10-06 09:49 NZDT.

Location: -36.8485,174.7635

Organisation: AS137409 GSL Networks Pty LTD

City: Auckland

Country: New Zealand

Region: Auckland

IP Address: 45.133.7.36

Timezone: Pacific /Auckland

Postal: 1010

Asn:

Route: 45.133.7.0/24

Type: Hosting

Asn: AS137409

Domain: Globalsecurelayer.com

Name: GSL Networks Pty LTD

Geolocation limitations: Geo-IP is approximate and may correspond to corporate outgoing traffic, proxies, or VPNs. It is not possible to determine a person's physical location with certainty based solely on this IP address without a court order or additional data from the internet service provider (ISP).

The evidence supports human access to our resource from the indicated IP address but does not allow us to determine a physical address or the exact location of the person.

Other details for police investigation

Useful visible elements:

1. Profile and cover photo(s): For recognition and OSINT comparisons; metadata if available, but FB usually cleans this up.
2. Aliases/name variants: In "About" section, under names, you can see his former name "Steven Austin" and his alias "Gouldyloxx." Researching these names in different OSINT tools did not yield any information relevant to this case.
3. Groups/pages followed: Groups related to Star Wars in Auckland. Useful for corroborating interests and circles.
4. Activity cadence: Dates and times of posts to estimate usual connection times.
5. Language and writing style: All posts are in English, useful for contextual attribution.

What we contribute:

- Message thread (includes Base64 and acceptance to receive details).
- Times and textual content of consent for event information (supports the thesis of voluntary interaction).

Limitation: Due to profile privacy, most of this data may not be accessible without cooperation from the platform or the owner; the police could request it through legal channels directly from Meta/Facebook.

Evidence of Criminal Wrongdoing and Applicable Laws

We limited our investigation to observational and passive data collection and used OSINT to search for any evidence. Using his IP address and Domain, we searched in '**VirusTotal**', '**Shodan**', '**Whois**', and '**abuseipdb**' and checked reputation of his IP and activities history, but no clear evidence of suspicious activity was found. See Appendix B to E for more details on investigation result.

Based on description of Christopher's conduct of unauthorised acecss of SecOp's confidential data, we have identified that he could have violated '**Crimes Act 1961**', **'Privacy Act 2020**', and '**Secret Commissions Act 1910**'. Punishment also applies if he used his work credentials to access data outside of his financial duties.

If he stole the company data and was planning to sell it to North Korean Government, this could mean he has violated **section 72** for attempting to commit a data theft crime, **section 78** for intention of communicating information with North Korea, **section 249** for accessing computer system for dishonest purpose, and possibly **section 252** if he accessed SecOp's computer system without authorisation under **Crimes Act 1961** (Parliamentary Counsel Office, 1961).

The **Privacy Act 2020** was also violated. Under the 13 principles of Privacy Act 2020, **principle 5** covers agencies must ensure that personal information is protected against loss, unauthorised access, use, modification, or disclosure and **principle 11** covers that personal information can only be disclosed in limited circumstances, such as with the person's consent or for the purpose it was obtained, or if required or permitted by law and Christopher accessed company's data without permission and his misconduct could cause SecOp to face heavy fines as he works for SecOp's financial department (Parliamentary Counsel Office, 2020).

Under **section 4** of **Secret Commissions Act 1910**, it is a criminal offence for an agent (such as an employee or representative) to accept bribes or gifts as an inducement or reward for doing something corruptly in relation to their principal's business (Parliamentary Counsel Office, 1910). If the suspect received any bribes from the North Korean Government to sell the company's data, this could mean he has violated **Secret Commissions Act 1910**.

Op-Sec Precautions

During our investigation, many OpSec precautions were applied to protect SecOps' systems, maintain evidence of integrity, and ensure compliance with legal and ethical standards.

Isolated environment

- All investigative traffic, the creation of the social media account, and the generation and testing of the Canarytoken were done through anonymisation services like VPN to prevent exposure to our IP.
- All work was conducted on a secure virtual machine (VM) instead of doing it on personal devices that could contain private data.
- This will prevent any accidental spread of malware on our devices.

Protection of Investigator Identity and Network

- When we created a Facebook account, we did not use our real account but created a fake sock puppet account that is not linked in any way to our personal identity, or SecOps to contact Christopher anonymously.
- The technique we used to embed a tracking link with Canarytoken was done passively. We only received data (IP address, User-Agent, time, geolocation) when the suspect voluntarily clicked the link.
- This protected our anonymity and prevented counter-reconnaissance.

Legal and ethical compliance

- We collected only necessary information (IP address, user-agent, geolocation) from CanaryToken. No personal or sensitive data unrelated to the case was accessed.
- We collected data of the suspect from OSINT using public legitimate websites such VirusTotal, Whois and Shodan.
- Any request for subscriber or provider logs was referred to the police, who hold the lawful authority to obtain such evidence.
- This ensures that our investigation aligns with the Privacy Act 2020.

<u>Secure communication and storage</u>

- All investigative notes and files were exchanged only over encrypted channels and stored in access-controlled folders.
- This safeguarded confidential information and preserved organisational reputation.

These precautions ensured that our investigation maintained confidentiality, integrity, and legality. By controlling access, verifying evidence of integrity, and following privacy and ethical standards, SecOps can protect both its infrastructure and the admissibility of the collected evidence.

Conclusion and Recommendations

The investigation successfully confirmed that the suspect, Christopher Haycox, accessed a monitoring link generated by Canarytoken that was embedded in a controlled social engineering message. The recorded hit provided a timestamp, IP address (45.133.7.36), and browser metadata originating from within the Auckland region, establishing a technical link between the suspect and the incident.

While no subset of confidential data was recovered from this investigation, the evidence indicates a deliberate attempt to engage with potentially stolen corporate material and communicate with an unauthorised third party.

Christopher's actions likely constitute breaches of the Crimes Act 1961 (sections 72, 78, 249, 252). He also infringed principles 5 and 11 of the Privacy Act 2020 through unauthorised disclosure of protected information. If he received money from the North Korean Government to sell the company's data, he is also accountable for breaching section 4 of the Secret Commissions Act 1910.

All evidence was collected lawfully with OSINT, maintaining full compliance with organisational policy and New Zealand legislation. While carrying out investigation, Op-Sec precautions must be taken. Investigation should be done in an isolated environment, investigator's own identity must be protected, comply with ethical and legal legislation, and investigative notes should be stored securely to maintain confidentiality, integrity, and legality.

References

Resources:

- https://www.canarytokens.org/nest/
- https://www.whois.com/
- https://www.virustotal.com/
- https://www.abuseipdb.com/
- https://www.shodan.io/

References:

Parliamentary Counsel Office. (1961, November 1). *Crimes Act 1961*. New Zealand Legislation. Retrieved from https://www.legislation.govt.nz/act/public/1961/0043/latest/dlm327382.html

Parliamentary Counsel Office. (2020, June 30). *Privacy Act 2020: Part 3: Information privacy principles and codes of practice*. New Zealand Legislation. Retrieved from https://www.legislation.govt.nz/act/public/2020/0031/latest/LMS23342.html

Parliamentary Counsel Office. (1910, December 3). *Secret Commissions Act 1910: Acceptance of such gifts by agent an offence*. New Zealand Legislation. Retrieved from https://www.legislation.govt.nz/act/public/1910/0040/latest/DLM177657.html

Appendices

Appendix A

Incident info

Date	IP	Channel
October 6, 2025 at 9:49:26 AM	**45.133.7.36**	**HTTP**

Basic info:

TOKEN TYPE
Web bug

INPUT CHANNEL
HTTP

×

USER-AGENT
Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/140.0.0.0 Safari/537.36

Time of hit: October 6, 2025 at 9:49:26 AM

Source IP: 45.133.7.36

Geo info:

LOCATION
-36.8485,174.7635

ORGANISATION
AS137409 GSL Networks Pty LTD

CITY
Auckland

COUNTRY
NZ

REGION

REGION
Auckland

IP
45.133.7.36

TIMEZONE
Pacific/Auckland

POSTAL
1010

Asn:

Asn:

ROUTE
45.133.7.0/24

TYPE
hosting

ASN
AS137409

DOMAIN
globalsecurelayer.com

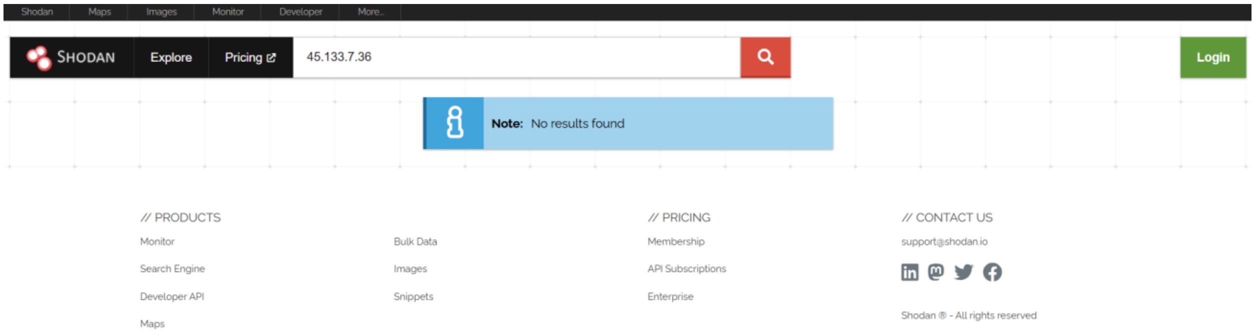
DOMAIN
globalsecurelayer.com

NAME
GSL Networks Pty LTD

Tor Known Exit Node: false

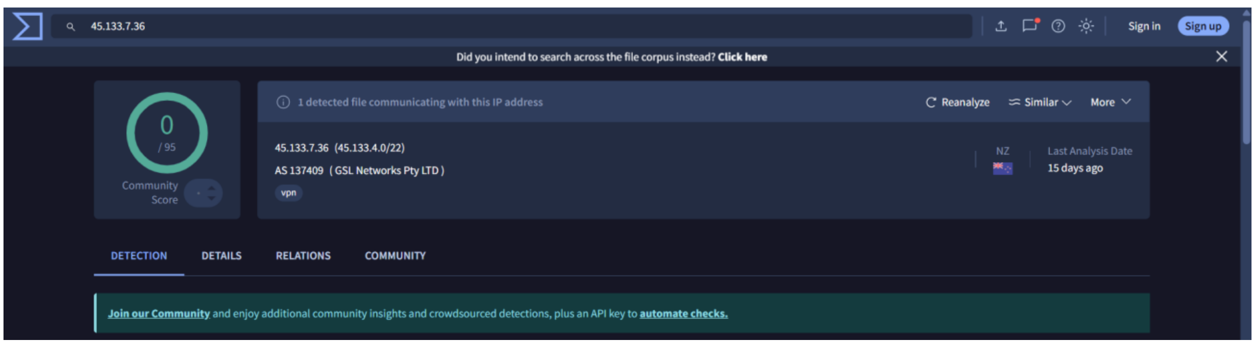
(CanaryTokens)

Appendix B



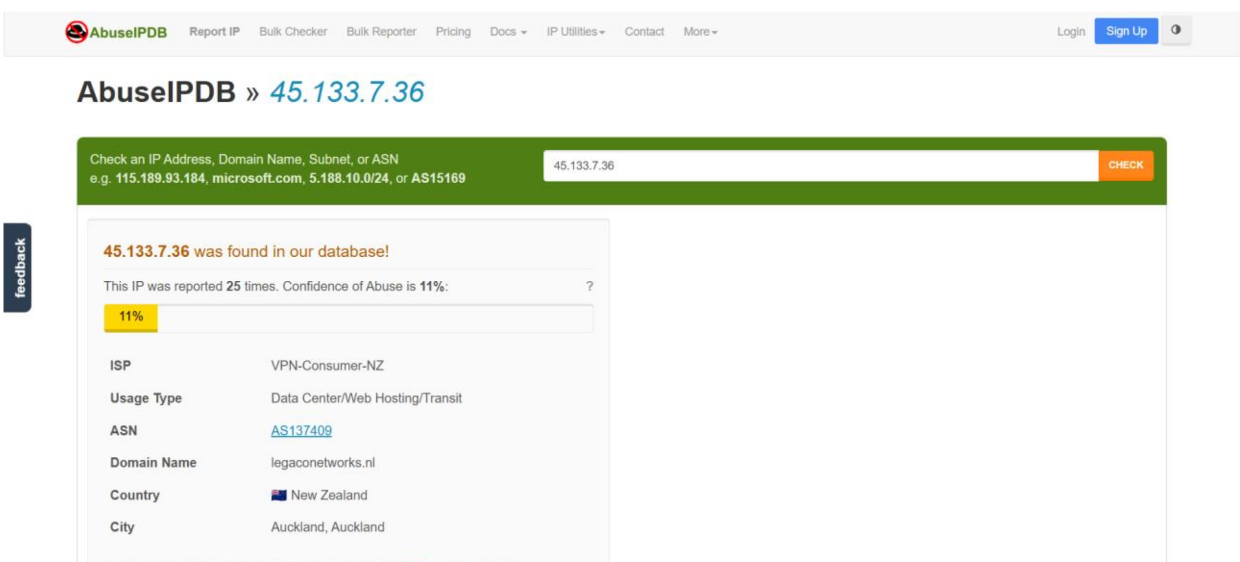
(Shodan)

Appendix C



(virustotal)

Appendix D



(abuseipdb)

Appendix E

Whois IP 45.133.7.36

Updated 1 day ago

```
% This is the RIPE Database query service.
% The objects are in RPSL format.
%
% The RIPE Database is subject to Terms and Conditions.
% See https://docs.db.ripe.net/terms-conditions.html

% Note: this output has been filtered.
%       To receive output for a database update, use the "-B" flag.

% Information related to '45.133.7.0 - 45.133.7.255'

% Abuse contact for '45.133.7.0 - 45.133.7.255' is 'abuse@legaconetworks.nl'

inetnum:        45.133.7.0 - 45.133.7.255
netname:        VCNZ-45-133-7-0
country:        NZ
geoloc:         -36.7502338 174.6941751
org:            ORG-VA29469-RIPE
admin-c:        LNBV
tech-c:         LNBV
status:         ASSIGNED PA
mnt-by:         PREFIXBROKER-MNT
created:        2020-03-24T10:25:35Z
last-modified:  2020-03-24T10:25:35Z
source:         RIPE
```

```
organisation:   ORG-VA29469-RIPE
org-name:       VPN-Consumer-NZ
org-type:       OTHER
address:        7a Parkhead Place
address:        Rosedale
address:        Auckland 0632
address:        New Zealand
abuse-c:        LNBV
mnt-ref:        PREFIXBROKER-MNT
mnt-by:         PREFIXBROKER-MNT
created:        2020-03-24T10:25:35Z
last-modified:  2020-03-24T10:25:35Z
source:         RIPE # Filtered

role:           Role object for Legaco Networks B.V.
address:        Kennedyplein 200
address:        5611ZT
address:        Eindhoven
address:        NETHERLANDS
phone:          +31403041481
abuse-mailbox:  abuse@legaconetworks.nl
admin-c:        JVV284-RIPE
admin-c:        WS4695-RIPE
nic-hdl:        LNBV
mnt-by:         nl-legaco-1-mnt
created:        2019-04-16T17:25:57Z
last-modified:  2019-04-16T17:29:29Z
source:         RIPE # Filtered
```

```
% Information related to '45.133.7.0/24AS137409'

route:          45.133.7.0/24
origin:         AS137409
mnt-by:         PREFIXBROKER-MNT
created:        2022-07-28T09:20:05Z
last-modified:  2022-07-28T09:20:05Z
source:         RIPE

% This query was served by the RIPE Database Query Service version 1.119 (BUSA)
```

(whois)